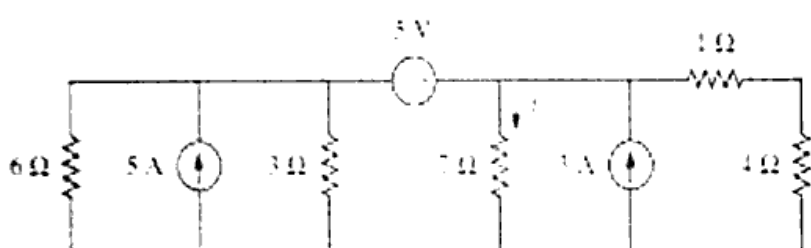
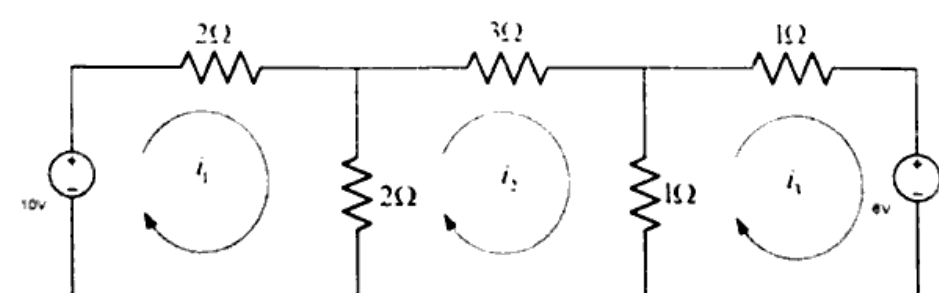
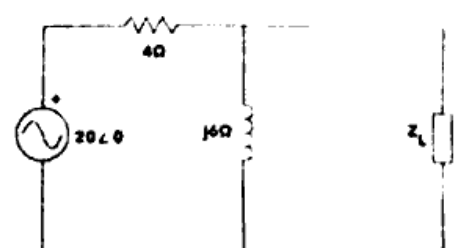
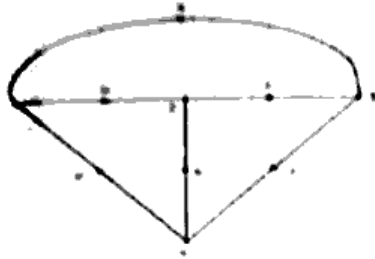


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Branch : Electrical Engineering Semester :IV Subject Code & Name: BTEEC401 Network Theory Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is relation between voltage, current and power?	CO1	6
B)	Solve the given network by using source transformation.	CO1	6
			
C)	What is meant by lumped and distributed parameters? Explain.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Apply Kirchhoff's voltage law on given network.	CO2	6
			
B)	Solve following network by using maximum power theorem on Z_L resistor.	CO2	6
			

C)	Find the 1. Incident matrix 2. Tie set matrix 3. Cut set matrix 	CO2	6
Q.3	Solve Any Two of the following.		12
A)	What is initial and final condition for transient response analysis in circuit.	CO3	6
B)	What is first order and second order differential equation of series R-L circuit.	CO3	6
C)	Explain various properties of inductor.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain the Impulse function & Ramp function with respect to Laplace Transform.	CO4	6
B)	Solve the following network to find Z parameter. 	CO4	6
C)	Derive an expression for (Hybrid) parameter H in terms of (Admittance) Y parameter.	CO4	6
Q.5	Solve Any Two of the following.		12
A)	Obtain an expression of the Frequency of resonance of a series and parallel resonating circuit.	CO5	6
B)	Explain low pass filter and high pass filter.	CO5	6
C)	The voltage across L and C at resonance is not the maximum voltage that can appear across it. Derive an expression for frequency at which the voltage across L and C are maximum.	CO5	6
	*** End ***		